

ORIGINAL ARTICLE

Blood Pressure Phenotypes and Cardio-Kidney Outcomes: Findings From UK Biobank Data

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BACKGROUND: Hypertension is a leading cause of cardio-kidney outcomes (CKO). However, distinct blood pressure (BP) phenotypes and their effect on CKO remain underexplored. This study aimed to evaluate the independent contributions of BP phenotypes to CKO.

METHODS: This prospective, population-based study included UK Biobank participants enrolled between 2006 and 2010 with baseline systolic BP ≥ 90 mm Hg and estimated glomerular filtration rate ≥ 60 mL/min per 1.73 m^2 , excluding those with prior chronic kidney disease and cardiovascular disease. Five BP phenotypes were constructed based on the 2017 American Heart Association hypertension criteria ($\geq 130/80$ mm Hg): normotension, systolic-diastolic hypertension, isolated systolic hypertension, isolated diastolic hypertension, and isolated low diastolic BP. The primary outcome was CKO, a composite of incident chronic kidney disease, 3-point major cardiovascular events, and all-cause mortality based on the *International Classification of Diseases* and Office of Population Censuses and Surveys Classification of Interventions and Procedures, Version 4, codes. Associations between BP phenotypes and CKO were evaluated using multivariable Cox proportional hazards models adjusted for demographic, lifestyle, and clinical covariates.

RESULTS: Among 322 328 participants (mean age, 55.8 ± 8.1 years; 42.5% men), 61 918 (19.2%) had normotension, 159 853 (49.6%) had systolic-diastolic hypertension, 43 939 (13.6%) had isolated systolic hypertension, 28 169 (8.7%) had isolated diastolic hypertension, and 28 339 (8.8%) had isolated low diastolic BP. Over a median follow-up of 13.6 years, 32 440 CKO events occurred. All BP phenotypes were associated with an increased CKO risk compared with normotension: isolated systolic hypertension (hazard ratio, 1.22 [95% CI, 1.17–1.27]), systolic-diastolic hypertension (hazard ratio, 1.21 [95% CI, 1.16–1.25]), isolated diastolic hypertension (hazard ratio, 1.11 [95% CI, 1.05–1.17]), and isolated low diastolic BP (hazard ratio, 1.10 [95% CI, 1.05–1.17]).

CONCLUSIONS: Distinct BP phenotypes beyond absolute BP thresholds confer differential risks for CKO. Interventional studies are needed to evaluate strategies targeting combined systolic-diastolic phenotypes to reduce CKO risk.

Key Words: blood pressure ■ humans ■ kidney ■ life style ■ phenotype

Hypertension is a well-established risk factor for both cardiovascular disease (CVD) and chronic kidney disease (CKD), with increasing global prevalence.^{1–3} Current clinical guidelines differ in diagnostic thresholds for hypertension, with the 2017 American College of Cardiology/American Heart Association (ACC/AHA) using a threshold of $\geq 130/80$ mm Hg, whereas the 2024 European Society of Cardiology (ESC) uses a higher threshold of $\geq 140/90$ mm Hg.^{2,3} These discrepant definitions result

in inconsistent classification of blood pressure (BP) phenotypes, such as isolated systolic hypertension (ISH) and isolated diastolic hypertension (IDH), which subsequently affect estimates of disease burden and risk prediction for CVD and CKD. Moreover, the long-term impact of guideline-defined BP phenotypes on cardio-kidney outcomes (CKO) remains poorly characterized.

Distinct BP phenotypes, such as systolic-diastolic hypertension (SDH), ISH, IDH, and isolated low

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WHAT IS KNOWN

- Hypertension is a major risk factor for both cardiovascular disease and chronic kidney disease.
- Current risk stratification primarily relies on absolute systolic and diastolic blood pressure thresholds.
- The combined impact of refined blood pressure phenotypes on integrated cardio-kidney outcomes has not been well characterized in large population-based cohorts.

WHAT THE STUDY ADDS

- Distinct blood pressure phenotypes are independently associated with differential risks of integrated cardio-kidney outcomes in a large prospective cohort.
- Isolated systolic hypertension confers increased risk for both cardiovascular and kidney outcomes, with effect sizes comparable to systolic-diastolic hypertension.
- Isolated low diastolic blood pressure is associated with elevated cardio-kidney risk, particularly among individuals with diabetes.
- The associations remain consistent when alternative hypertension thresholds (2024 European Society of Cardiology criteria) are applied.
- These findings suggest that blood pressure phenotype-based classification may enhance risk stratification beyond conventional blood pressure thresholds.

Nonstandard Abbreviations and Acronyms

3P-MACE	3-point major adverse cardiovascular event
ACC	American College of Cardiology
AHA	American Heart Association
BMI	body mass index
BP	blood pressure
CKD	chronic kidney disease
CKO	cardio-kidney outcomes
CVD	cardiovascular disease
DBP	diastolic blood pressure
ESC	European Society of Cardiology
HR	hazard ratio
IDH	isolated diastolic hypertension
ILDBP	isolated low diastolic blood pressure
ISH	isolated systolic hypertension
SBP	systolic blood pressure
SDH	systolic-diastolic hypertension

diastolic BP (ILDBP), reflect different hemodynamic mechanisms and clinical risk profiles. ISH, which is prevalent among older adults and typically reflects

increased arterial stiffness, has been consistently associated with a high risk of CVD.^{4–8} IDH, which is more prevalent in younger individuals, has traditionally been considered relatively benign. However, emerging evidence suggests that IDH may be associated with increased cardiovascular risk in certain populations.^{9,10} Although findings have been inconsistent, likely due to heterogeneity in BP, IDH has been associated with impaired coronary perfusion and a high risk of adverse outcomes, particularly in individuals with diabetes or atherosclerosis. While intensive systolic BP (SBP)-lowering has demonstrated cardiovascular benefits following the SPRINT trial, the safety of lowering diastolic BP (DBP) remains uncertain, particularly in high-risk populations. Observational studies suggest a J-shaped association between DBP and adverse outcomes; however, guidelines provide limited recommendations regarding lower DBP thresholds.^{11–13} In addition, the kidney-specific consequences of BP phenotypes remain understudied, despite the interdependence of cardiac and renal systems.

Given these knowledge gaps, this study aimed to evaluate the association between BP phenotypes and CKO risk, including incident CKD and major adverse cardiovascular events, using data from the UK Biobank, a large prospective population-based cohort.

METHODS

Data Availability

Data supporting the findings of this study are available from the corresponding author upon reasonable request.

Study Population and Data Source

This study utilized data from the UK Biobank, a large prospective cohort study that recruited over 500 000 participants aged 40 to 69 years between 2006 and 2010. Detailed descriptions of the UK Biobank protocol have been published previously.¹⁴ At baseline, information on demographic and socioeconomic status, including age, sex, education and income levels, smoking status, alcohol consumption, and medical history, was collected. Participants self-reported their race and ethnicity as Asian, Black, White, or Multiethnic/other. Physical examinations were conducted to obtain anthropometric data and vital sign measurements, and blood and urine samples were collected for laboratory testing. The estimated glomerular filtration rate was calculated using the CKD Epidemiology Collaboration equation.¹⁵ Participants were prospectively followed up for incident health outcomes through national health records and follow-up study visits. For this study, follow-up was conducted through April 13, 2021. Individuals in the UK Biobank cohort with missing baseline data for serum creatinine or BP were excluded. Participants with a previous diagnosis of CKD, CVD, or cancer were also excluded. Additional exclusions were made for individuals with an estimated glomerular filtration rate <60 mL/min per 1.73 m² or an SBP of <90 mm Hg at baseline. After applying these criteria, 322 218 participants were included in the final analysis (Figure 1). This study was conducted under UK

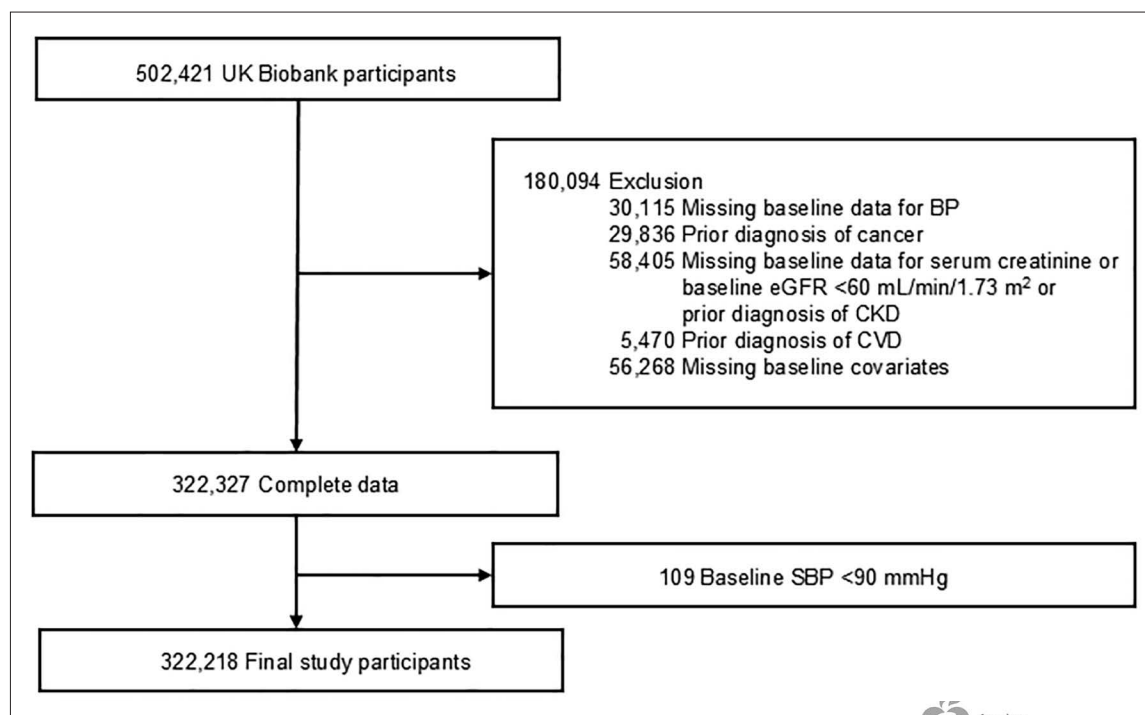


Figure 1. Flow of study participants.

The flowchart illustrates the selection of study participants from the UK Biobank. Participants with missing blood pressure (BP), serum creatinine measurements, or baseline covariates, preexisting chronic kidney disease (CKD), cardiovascular disease (CVD), estimated glomerular filtration rate (eGFR) <60 mL/min per 1.73 m², or systolic BP (SBP) <90 mm Hg were excluded. A total of 322 218 participants were included in the final analysis.

Biobank application number 72527. The UK Biobank received ethical approval from the Northwest Multicenter Research Ethics Committee. The Massachusetts General Hospital institutional review board approved the secondary analysis of these data. All participants provided electronically signed informed consent.

BP Measurements and Classification of BP Phenotypes

Following 5 minutes of seated rest, BP was measured twice consecutively at 1-minute intervals using an Omron 705 IT electronic monitor (OMRON Healthcare), with the average of the 2 automated readings used for data analysis. For this analysis, antihypertensive medications were grouped into 4 major classes: renin-angiotensin system inhibitors (angiotensin-converting enzyme inhibitors or angiotensin receptor blockers), β -blockers, calcium channel blockers, and diuretics. The number of classes of antihypertensive agents taken by each participant was determined. Participants were classified into 5 BP phenotypes based on their baseline SBP and DBP following the 2017 AHA guidelines for the management of elevated BP and hypertension, which define hypertension as SBP $\geq$ 130 mmHg and DBP $\geq$ 80 mmHg.¹⁶ The groups were as follows: (1) SDH, with SBP $\geq$ 130 mmHg and DBP $\geq$ 80 mmHg; (2) ISH, with SBP $\geq$ 130 mmHg and DBP <80 mmHg; (3) IDH, with SBP <130 mmHg and DBP $\geq$ 80 mmHg; (4) normotension (reference group), with SBP <130 mmHg and 70 $\leq$ DBP <80 mmHg; and (5) ILDBP, with SBP <130 mmHg and DBP <70 mmHg.

Cardio-Kidney Outcomes

The primary outcome was a composite of the development of CKD, 3-point major adverse cardiovascular events (3P-MACE), and all-cause mortality, ascertained by the Hospital Episode Statistics data. CKD development was defined as the first occurrence of diagnostic codes from the *International Classification of Diseases*, Tenth Revision, or procedure codes from the Office of Population Censuses and Surveys Classification of Interventions and Procedures, Version 4, including kidney transplantation, dialysis, and vascular access procedures indicative of advanced CKD. The 3P-MACE included ischemic stroke, myocardial infarction, and cardiovascular death based on *International Classification of Diseases*, Ninth Revision, and *International Classification of Diseases*, Tenth Revision codes. Detailed outcome definitions and codes are provided in [Table S1](#). Follow-up was calculated from the baseline BP measurement to the earliest of CKD or 3P-MACE diagnosis, death, or the end of follow-up.

Statistical Analysis

Continuous variables were presented as means $\pm$ SD, while categorical variables were expressed as frequencies and percentages. Normality of the data distribution was assessed using the Kolmogorov-Smirnov test before statistical analysis. For between-group comparisons, 1-way ANOVA was applied to normally distributed continuous variables, and the χ^2 or Fisher exact test was used for categorical variables. Nonnormally distributed data were reported as medians with interquartile ranges and were compared using the Kruskal-Wallis test.

Kaplan-Meier analysis with the log-rank test was used to assess differences in the cumulative incidence of study outcomes among the isolated BP groups. Survival time was defined as the period from the baseline visit to the occurrence of the study outcome or last follow-up. Participants who were lost to follow-up or who died were censored at the last known date of contact. To examine the threshold BP levels associated with CKO, we assessed the associations between SBP and DBP categories and CKO using multivariable Cox proportional hazards models. To evaluate the independent predictive value of BP phenotypes for CKO, we developed multivariable cause-specific Cox proportional hazards models that accounted for the competing risk of death by censoring individuals at the time of death if it occurred before CKD. Model 1 was unadjusted; Model 2 was adjusted for age and sex; and Model 3 included additional adjustments for body mass index (BMI), smoking status, alcohol consumption, use of antihypertensive medications (renin-angiotensin system inhibitors, β -blockers, calcium channel blockers, and diuretics), HDL-C (high-density lipoprotein cholesterol), LDL-C (low-density lipoprotein cholesterol), baseline estimated glomerular filtration rate, and glycated hemoglobin. This approach estimates the hazard of the outcome among individuals who remain alive and event-free.¹⁷ Given the observational design, all analyses were restricted to participants with complete data on exposure, outcomes, and covariates included in the multivariable models. Proportional hazards assumptions were confirmed using Schoenfeld residuals. Secondary analyses were conducted to test the robustness of the association between BP phenotypes and incident CKO. First, hypertension was redefined using the 2024 ESC guideline (SBP ≥ 140 mmHg and DBP ≥ 90 mmHg). Under this definition, the BP phenotypes were as follows: (1) SDH, defined as SBP ≥ 140 mmHg and DBP ≥ 90 mmHg; (2) ISH, defined as SBP ≥ 140 mmHg and DBP < 90 mmHg; (3) IDH, defined as SBP < 140 mmHg and DBP ≥ 90 mmHg; (4) normotension, defined as SBP < 140 mmHg and 70 $\leq$ DBP < 90 mmHg; and (5) ILDBP, defined as SBP < 140 mmHg and DBP < 70 mmHg. Second, ILDBP was redefined using a DBP threshold of < 60 mmHg, and the main analyses were repeated. Third, social determinants of health-related variables were additionally incorporated into the final Cox model. Fourth, competing risk analyses using Fine-Gray subdistribution hazard models were additionally performed to account for death as a competing event for kidney outcomes. Fifth, the association between BP phenotypes and the primary outcome was assessed among individuals who were naive to antihypertensive agents. Finally, subgroup analyses were performed according to age, sex, BMI, and diabetes status. Statistical significance was set at $P < 0.05$. All statistical analyses were performed using R software (version 4.2.2; <http://www.R-project.org>).

RESULTS

Baseline Characteristics

The baseline characteristics of the study population are presented in Table 1. The mean age of the participants was 55.8 ± 8.1 years, and 42.5% were men. The distribution of the BP phenotypes among the study population is shown in Figure 2. The average SBP and DBP values

in each BP pattern group were as follows: in the normotension, SDH, ISH, IDH, and ILDBP groups, SBP values were 119.8 ± 6.4 , 149.9 ± 14.7 , 141.1 ± 10.3 , 124.3 ± 4.3 , and 112.7 ± 8.5 mmHg, respectively, while DBP values were 74.8 ± 2.8 , 89.8 ± 7.1 , 75.0 ± 4.0 , 83.7 ± 3.3 , and 65.5 ± 3.5 mmHg, respectively. Participants in the ISH group were the oldest, whereas those in the IDH group were the youngest. The proportion of men was highest in the SDH group and lowest in the IDH group. Regarding ethnicity, the ISH group had the highest proportion of White participants, whereas the IDH group had a higher representation of Black and Asian participants. BMI was highest in the SDH group and lowest in the ILDBP group. The prevalence of current smokers was highest in the IDH group, whereas that of never smokers was highest in the normotension group. Current alcohol consumption was most common in the SDH group, whereas the proportion of never drinkers was highest in the ILDBP group. The physical activity level was lowest in the IDH group. The proportion of participants with higher education was greatest in the ILDBP group, whereas those with lower education levels were most common in the IDH group. Income levels were highest in the normotension and ILDBP groups and lowest in the ISH group. Overall, 17.6% of the study population was receiving treatment for hypertension using antihypertensive medications. The ISH group had the highest usage rates across all antihypertensive drug classes, including angiotensin-converting enzyme inhibitors/angiotensin II receptor blockers, β -blockers, calcium channel blockers, and diuretics. A history of diabetes was most prevalent in the ISH group and least prevalent in the normotension and ILDBP groups. Laboratory findings showed that the serum albumin level was lowest in the ILDBP group. The estimated glomerular filtration rate level was highest in the ILDBP group and lowest in the SDH group. LDL-C and triglyceride levels were lowest in the ILDBP group, while HDL-C levels were highest in the ILDBP group. Fasting glucose and glycated hemoglobin levels were lowest in the ILDBP group and highest in the ISH group.

Risk of CKO by SBP and DBP Categories

To explore optimal BP thresholds associated with CKO risk, multivariable Cox proportional hazards models were constructed and stratified by SBP (< 110 , 110–119 [Reference], 120–129, 130–139, 140–149, 150–159, and ≥ 160 mmHg) and DBP (< 60 , 60–69, 70–79 [Reference], 80–89, and ≥ 90 mmHg) categories (Table 2). For SBP categories, CKO risk was significantly elevated at < 110 mmHg and increased progressively across higher SBP categories above 130 mmHg, demonstrating a graded association. Similarly, for DBP categories, the highest risk of CKO was observed in the < 60 mmHg group; DBP groups with 60–69, 80–89, and ≥ 90 mmHg

Table 1. Baseline Characteristics

	BP phenotypes					P value
	Normotension (n=61 918)	SDH (n=159 853)	ISH (n=43 939)	IDH (n=28 169)	ILDBP (n=28 339)	
Demographic data						
Age, y	53.0±7.9	57.0±7.7	59.7±7.3	52.1±7.4	53.7±8.3	<0.001
Sex, men, n (%)	20 608 (33.3%)	81 240 (50.8%)	17 520 (39.9%)	10 496 (37.3%)	7 162 (25.3%)	<0.001
Ethnic background, n (%)						<0.001
White	57 950 (93.6%)	151 341 (94.7%)	42 307 (96.3%)	25 967 (92.2%)	26 755 (94.4%)	
Black	842 (1.4%)	2290 (1.4%)	365 (0.8%)	526 (1.9%)	276 (1.0%)	
Asian	1729 (2.8%)	3513 (2.2%)	690 (1.6%)	985 (3.5%)	660 (2.3%)	
Mixed	507 (0.8%)	836 (0.5%)	181 (0.4%)	215 (0.8%)	221 (0.8%)	
Others	685 (1.1%)	1277 (0.8%)	254 (0.6%)	354 (1.3%)	324 (1.1%)	
BMI, kg/m²	26.0±4.2	28.3±4.8	26.6±4.3	28.1±5.1	24.4±3.7	<0.001
Smoking status, n (%)						<0.001
Current	7334 (11.8%)	15 687 (9.8%)	4169 (9.5%)	3479 (12.4%)	3481 (12.3%)	
Ex-smoker	18 742 (30.3%)	56 053 (35.1%)	15 782 (35.9%)	8512 (30.2%)	8760 (30.9%)	
Never	35 631 (57.5%)	87 512 (54.7%)	23 815 (54.2%)	16 064 (57.0%)	15 998 (56.5%)	
Alcohol status, n (%)						<0.001
Current	56 897 (91.9%)	148 443 (92.9%)	40 163 (91.4%)	25 770 (91.5%)	25 744 (90.8%)	
Ex-drinker	2228 (3.6%)	4794 (3.0%)	1615 (3.7%)	1039 (3.7%)	1470 (4.1%)	
Never	2709 (4.4%)	6396 (4.0%)	2121 (4.8%)	1314 (4.7%)	1379 (4.9%)	
Physical activity (Mets)	420.0 (120.0–1120.0)	480.0 (120.0–1200.0)	540.0 (160.0–1440.0)	360.0 (80.0–960.0)	480.0 (160.0–1200.0)	<0.001
SBP, mm Hg	119.8±6.4	149.9±14.7	141.1±10.3	124.3±4.3	112.7±8.5	<0.001
DBP, mm Hg	74.8±2.8	89.8±7.1	75.0±4.0	83.7±3.3	65.5±3.5	<0.001
Education, n (%)						<0.001
High	31 702 (51.2%)	66 116 (41.4%)	17 235 (39.2%)	13 467 (47.8%)	14 699 (51.9%)	
Middle	9732 (15.7%)	29 002 (18.1%)	7316 (16.7%)	4949 (17.6%)	4263 (15.0%)	
Low	13 282 (21.5%)	34 998 (21.9%)	9519 (21.7%)	6394 (22.7%)	5911 (20.9%)	
Income£, n (%)						<0.001
>100 000	4182 (6.8%)	6689 (4.2%)	1483 (3.4%)	1648 (5.9%)	1929 (6.8%)	
52 000–100 000	13 456 (21.8%)	27 740 (17.4%)	5987 (13.7%)	6207 (22.1%)	5678 (20.1%)	
31 000–51 999	14 898 (24.1%)	36 341 (22.8%)	8861 (20.2%)	7069 (25.1%)	6592 (23.3%)	
18 000–30 999	12 053 (19.5%)	35 475 (22.2%)	10 323 (23.6%)	5358 (19.1%)	5541 (19.6%)	
<18 000	9612 (15.5%)	30 190 (18.9%)	9676 (22.1%)	4444 (15.8%)	4834 (17.1%)	
Townsend deprivation index	−2.1 (−3.6 to 0.6)	−2.2 (−3.7 to 0.3)	−2.3 (−3.7 to 0.1)	−1.9 (−3.5 to 1.0)	−2.1 (−3.6 to 0.7)	<0.001
Antihypertensive medications, n (%)						<0.001
ACE inhibitor/ARB	2193 (3.5%)	14 658 (9.2%)	4136 (9.4%)	1553 (5.5%)	858 (3.0%)	
β-Blocker	2133 (3.4%)	7510 (4.7%)	3407 (7.8%)	859 (3.0%)	1363 (4.8%)	
Calcium channel blocker	1459 (2.4%)	12 119 (7.6%)	3875 (8.8%)	981 (3.5%)	596 (2.1%)	
Diuretics	1817 (2.9%)	11 643 (7.3%)	3214 (7.3%)	1268 (4.5%)	733 (2.6%)	
Comorbidities, n (%)						<0.001
Hypertension	7230 (11.8%)	137 252 (86.3%)	28 589 (65.4%)	7020 (25.2%)	2911 (10.3%)	
Diabetes	2667 (4.3%)	10 974 (6.9%)	4269 (9.7%)	1423 (5.1%)	1209 (4.3%)	
Laboratory data						
Albumin, g/L	4.5±0.3	4.5±0.3	4.5±0.3	4.5±0.3	4.4±0.3	<0.001
eGFR, mL/min per 1.73 m²	90.3±14.4	85.4±13.9	85.9±13.3	89.6±14.7	91.7±14.2	<0.001
LDL-C, mg/dL	135.3±30.9	143.1±34.8	135.3±34.8	139.2±30.9	131.5±30.9	<0.001
HDL-C, mg/dL	58.0±15.5	54.1±15.5	58.0±15.5	54.1±15.5	1.5±0.4	<0.001

(Continued)

Table 1. Continued

	BP phenotypes					P value
	Normotension (n=61 918)	SDH (n=159 853)	ISH (n=439 339)	IDH (n=28 169)	ILDBP (n=28 339)	
Triglyceride, mg/dL	132.9±79.7	168.3±97.4	150.6±88.6	150.6±88.6	123.9±70.9	<0.001
Fasting glucose, mg/dL	88.3±18.0	93.7±21.6	95.5±25.2	90.1±19.8	88.3±18.0	<0.001
HbA1c, %	5.3±0.5	5.5±0.6	5.5±0.6	5.4±0.6	5.3±0.5	<0.001

Note: Data are presented as mean (SD), median (IQR), or number (%). Physical activity was assessed using metabolic equivalent task-based activity measures derived from self-reported physical activity questionnaires. Education level was categorized according to the UK educational system (high=university degree, A/AS levels; middle=O levels/GCSEs; low=CSEs, NVQ, HND, HNC, or other professional qualifications). Townsend deprivation index is an area-level measure of socioeconomic deprivation, with lower values indicating lower deprivation and higher socioeconomic status. ACE indicates angiotensin-converting enzyme; ARB, angiotensin II receptor blocker; BMI, body mass index; BP, blood pressure; CRP, C-reactive protein; eGFR, estimated glomerular filtration rate; HbA1c, glycated hemoglobin; HDL-C, high-density lipoprotein cholesterol; HOMA-IR, Homeostatic Model Assessment for Insulin Resistance; IDH, isolated diastolic hypertension; ILDBP, isolated low diastolic blood pressure; IQR, interquartile range; ISH, isolated systolic hypertension; LDL-C, low-density lipoprotein cholesterol; MF-ratio, muscle to fat ratio; SBP, systolic blood pressure; and SDH, systolic and diastolic hypertension.

were also associated with greater risk than the reference range of 70–79 mm Hg.

Association Between BP Phenotypes and CKO

During a median follow-up of 13.6 (12.8–14.3) years, 32 440 CKO events occurred (Table 3). The incidence rate of CKO was highest in the ISH group, with rates of 10.6, 8.9, 5.4, 5.4, and 4.8 per 1000 person-years in the ISH, SDH, ILDBP, IDH, and normotension groups, respectively ($P<0.001$). The cumulative incidence of CKO was significantly higher across all BP pattern groups than that in the normotension group, with the ISH group showing the highest incidence (Figure 3A). In the fully adjusted multivariable Cox proportional hazards model, all BP pattern groups were associated with a significantly increased risk of CKO. The highest risk was observed in the ISH group (hazard ratio [HR], 1.22 [95% CI, 1.17–1.27]), followed by the SDH (HR, 1.21 [95% CI, 1.16–1.25]), IDH (HR, 1.11 [95% CI, 1.05–1.17]), and

ILDBP (HR, 1.10 [95% CI, 1.05–1.17]) groups, compared with the normotension group.

For kidney-specific outcomes, 6926 cases of incident CKD were observed during a median follow-up of 13.6 (12.9–14.3) years (Table 3). The ISH group had the highest incidence rate (2.5 per 1000 person-years), followed by the SDH, IDH, ILDBP, and normotension groups (1.8, 1.2, 1.1, and 0.9 per 1000 person-years; $P<0.001$). The cumulative incidence of kidney outcomes was significantly higher in the SDH, ISH, and IDH groups than that in the normotension group (Figure 3B). Fully adjusted multivariable Cox models showed increased risks of kidney outcomes in both the ISH (HR, 1.31 [95% CI, 1.20–1.43]) and SDH (HR, 1.22 [95% CI, 1.13–1.31]) groups.

Regarding 3P-MACE, 14 157 3P-MACE events occurred during a median follow-up of 13.6 (12.8–14.3) years (Table 3). The highest incidence rate was observed in the ISH group (5.2 per 1000 person-years), followed by the SDH, IDH, ILDBP, and normotension groups (4.6, 2.5, 2.4, and 2.1 per 1000 person-years;

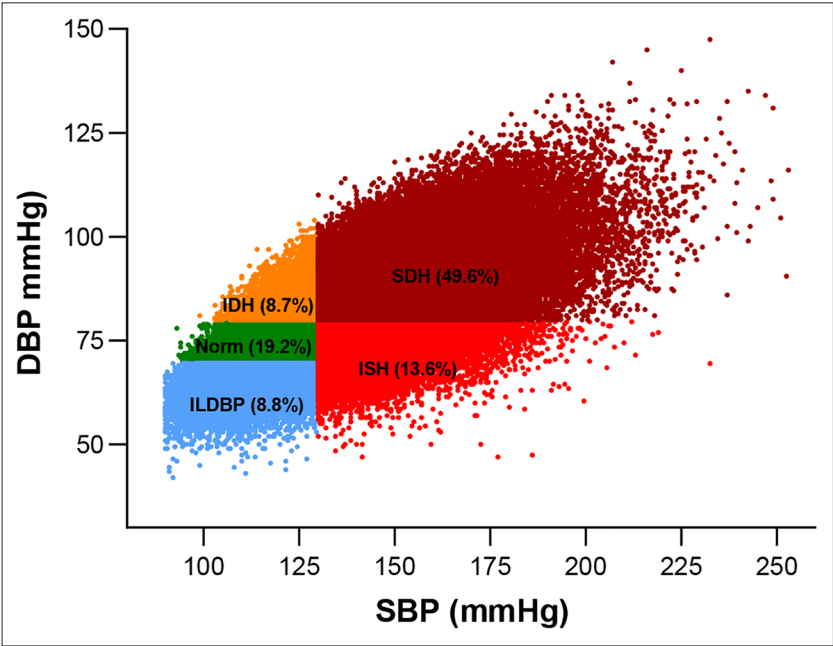


Figure 2. Distribution of blood pressure (BP) phenotypes. Scatter plot showing the distribution of systolic BP (SBP) and diastolic BP (DBP) among study participants. BP phenotypes were classified using the 2017 American College of Cardiology/American Heart Association criteria: normotension (Norm; <130/<80 mmHg), systolic and diastolic hypertension (SDH; ≥130/≥80 mmHg), isolated systolic hypertension (ISH; ≥130/<80 mmHg), isolated diastolic hypertension (IDH; <130/≥80 mmHg), and isolated low diastolic BP (ILDBP; <130/<70 mmHg). Values within each region indicate the proportion of participants in each phenotype group.

Table 2. Risk of Cardio-Kidney Outcomes by SBP and DBP Categories

	Events	1000-PYs	Rate	Crude HR (95% CI)	Adjusted* HR (95% CI)
SBP, mm Hg					
<110	841	207.4	4.05	0.91 (0.84–0.98)	1.09 (1.01–1.18)
110–119 (Reference)	2318	518.3	4.47	1 (Reference)	1 (Reference)
120–129	4864	847.8	5.74	1.28 (1.22–1.35)	1.04 (0.99–1.09)
130–139	6568	918.4	7.15	1.60 (1.53–1.68)	1.09 (1.03–1.14)
140–149	6579	745.9	8.82	1.98 (1.89–2.07)	1.16 (1.10–1.21)
150–159	5124	491.4	10.43	2.34 (2.23–2.46)	1.24 (1.18–1.30)
≥ 160	6146	487.1	12.62	2.84 (2.71–2.98)	1.38 (1.31–1.44)
DBP, mm Hg					
<60	238	31	7.68	1.12 (0.98–1.27)	1.20 (1.05–1.37)
60–69	2621	402.7	6.51	0.95 (0.91–0.99)	1.05 (1.00–1.09)
70–79 (Reference)	9131	1329.3	6.87	1 (Reference)	1 (Reference)
80–89	11 911	1518.3	7.85	1.14 (1.11–1.18)	1.03 (1.00–1.06)
≥ 90	8539	934.9	9.13	1.34 (1.30–1.38)	1.17 (1.14–1.21)

BMI indicates body mass index; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; HbA1c, glycated hemoglobin; HDL-C, high-density lipoprotein cholesterol; HR, hazard ratio; LDL-C, low-density lipoprotein cholesterol; PY, person-years; and SBP, systolic blood pressure.

*Adjusted for age, sex, smoking, and alcohol status, BMI, use of antihypertensive medications, baseline eGFR, HDL-C, LDL-C, and HbA1c.



$P < 0.001$). The cumulative incidence of 3P-MACE significantly increased in all BP groups, with the ISH group showing the highest incidence (Figure 3C). In the fully adjusted Cox model, all BP patterns were associated with a higher risk of 3P-MACE than that in the normotension group. The SDH group showed the greatest risk (HR, 1.37 [95% CI, 1.30–1.44]), followed by the ISH (HR, 1.36 [95% CI, 1.28–1.45]), ILDBP (HR, 1.14 [95% CI, 1.05–1.24]), and IDH (HR, 1.13 [95% CI, 1.05–1.23]) groups.

Secondary Analyses

Several secondary analyses were conducted to evaluate the robustness of our primary findings. First, to examine whether reclassification of BP patterns using alternative thresholds would alter the associations with outcomes, we applied the 2024 ESC hypertension criteria (SBP ≥ 140 mmHg and DBP ≥ 90 mmHg). The resulting distribution of BP phenotypes is shown in Figure S1. Using these thresholds, all BP pattern groups remained significantly associated with an increased risk of CKO after full adjustment for covariates. For kidney-specific outcomes and 3P-MACE, the findings were consistent with the primary analyses: SDH, ISH, and ILDBP were associated with an increased risk of kidney outcomes, and all BP phenotypes were associated with an increased risk of 3P-MACE (Table S2). Second, when the ILDBP definition threshold was lowered to DBP < 60 mmHg, consistent associations with CKO were observed, despite the relatively small number of participants classified in this group ($n=2198$; Table S3).

Third, after additional adjustment for social determinants of health-related variables, the associations between BP phenotypes and CKO risk remained consistent (Table S4). Fourth, using Fine-Gray subdistribution hazard models that accounted for death as a competing event for kidney outcomes, the overall associations remained materially unchanged (Table S5). The direction and magnitude of the associations were generally consistent with those observed in the primary cause-specific Cox proportional hazards analyses, supporting the robustness of the main findings after consideration of competing risks. Fifth, the association between BP phenotypes and CKO risk was evaluated in participants who were naïve to antihypertensive medication. The associations differed significantly according to baseline antihypertensive treatment status (P for interaction < 0.001). Among untreated participants, the SDH (HR, 1.61 [95% CI, 1.51–1.71]), ISH (HR, 1.48 [95% CI, 1.38–1.60]), and IDH (HR, 1.26 [95% CI, 1.14–1.38]) groups were associated with an increased risk of CKO, unlike the ILDBP group (Table S6). Finally, we conducted stratified subgroup analyses to investigate potential effect modification in the association between BP patterns and CKO by age (< 60 versus $60–69$ and ≥ 70 years), sex, BMI (< 25 versus ≥ 25 kg/m²), and diabetes (Table S7). Among participants aged < 60 years, SDH, ISH, and IDH were associated with an increased CKO risk, while ILDBP was not. However, in participants aged ≥ 60 years, SDH, ISH, and ILDBP were associated with greater risk, whereas IDH was not, with these associations appearing more pronounced among those aged ≥ 70 years (P for interaction < 0.001). Furthermore, all BP pattern groups were

Table 3. Association Between BP Phenotypes and Cardio-Kidney Outcomes

	Events	1000-PYs	Rate	Model 1	Model 2	Model 3
				HR (95% CI)	HR (95% CI)	HR (95% CI)
CKO						
Normotension	3971	823.8	4.82	1.00 (Reference)	1.00 (Reference)	1.00 (Reference)
SDH	18 440	2078.3	8.87	1.85 (1.79–1.91)	1.26 (1.22–1.31)	1.21 (1.16–1.25)
ISH	5977	564.4	10.59	2.20 (2.12–2.29)	1.30 (1.25–1.35)	1.22 (1.17–1.27)
IDH	2010	374.9	5.36	1.12 (1.06–1.18)	1.21 (1.14–1.27)	1.11 (1.05–1.17)
ILDBP	2042	374.7	5.45	1.13 (1.07–1.19)	1.09 (1.04–1.15)	1.10 (1.05–1.17)
Kidney outcome						
Normotension	826	832.5	0.99	1.00 (Reference)	1.00 (Reference)	1.00 (Reference)
SDH	3794	2124.9	1.79	1.81 (1.68–1.95)	1.32 (1.22–1.42)	1.22 (1.13–1.31)
ISH	1445	577.6	2.5	2.54 (2.33–2.76)	1.40 (1.28–1.52)	1.31 (1.20–1.43)
IDH	435	379.5	1.15	1.16 (1.03–1.30)	1.33 (1.18–1.49)	1.09 (0.97–1.23)
ILDBP	426	378.9	1.12	1.13 (1.01–1.27)	1.01 (0.90–1.14)	1.12 (0.99–1.26)
3P-MACE						
Normotension	1757	827.2	2.12	1.00 (Reference)	1.00 (Reference)	1.00 (Reference)
SDH	9614	2091.8	4.6	2.17 (2.06–2.28)	1.44 (1.37–1.52)	1.37 (1.30–1.44)
ISH	2938	569.3	5.16	2.43 (2.29–2.58)	1.48 (1.40–1.57)	1.36 (1.28–1.45)
IDH	939	376.6	2.49	1.18 (1.09–1.27)	1.24 (1.15–1.34)	1.13 (1.05–1.23)
ILDBP	909	376.3	2.42	1.14 (1.05–1.23)	1.14 (1.05–1.24)	1.14 (1.05–1.24)

Five BP phenotypes were constructed based on the 2017 American Heart Association hypertension criteria ($\geq 130/80$ mmHg): (1) SDH, with SBP ≥ 130 mmHg and DBP ≥ 80 mmHg; (2) ISH, with SBP ≥ 130 mmHg and DBP < 80 mmHg; (3) IDH, with SBP < 130 mmHg and DBP ≥ 80 mmHg; (4) normotension (reference group), with SBP < 130 mmHg and DBP < 80 mmHg; and (5) ILDBP, with SBP < 130 mmHg and DBP < 70 mmHg. Model 1: unadjusted. Model 2: Model 1 + age, sex. Model 3: Model 2 + smoking and alcohol status, BMI, use of antihypertensive medications, baseline eGFR, HDL-C, LDL-C, and HbA1c. 3P-MACE indicates 3-point major adverse cardiovascular events; BMI, body mass index; BP, blood pressure; CKO, cardio-kidney outcomes; DBP, diastolic blood pressure; eGFR, estimated glomerular filtration rate; HbA1c, glycated hemoglobin; HDL-C, high-density lipoprotein cholesterol; HR, hazard ratio; IDH, isolated diastolic hypertension; ILDBP, isolated low diastolic blood pressure; ISH, isolated systolic hypertension; LDL-C, low-density lipoprotein cholesterol; PY, person-years; SBP, systolic blood pressure; and SDH, systolic and diastolic hypertension.

associated with an elevated overall CKO risk regardless of sex ($P=0.222$). When stratified by BMI, SDH, ISH, and IDH were associated with an increased CKO risk in participants with BMI < 25 kg/m², and in participants with BMI ≥ 25 kg/m², SDH, ISH, and ILDBP were associated with an increased risk (P for interaction < 0.001). Among participants without diabetes, all BP phenotypes were significantly associated with an increased CKO risk. However, among patients with diabetes, the ISH and ILDBP groups were associated with an increased risk of CKO (P for interaction < 0.001). This association was consistent across both kidney outcomes and 3P-MACE, with the ILDBP group showing a particularly high risk among individuals with diabetes (Tables S8 and S9).

DISCUSSION

In this large-scale prospective cohort study using UK Biobank data, we found independent and variable associations between each BP phenotype, classified based on the 2017 AHA hypertension criteria ($\geq 130/80$ mmHg), and CKO risk, including incident CKD and 3P-MACE. We observed that SDH, ISH, IDH, and ILDBP were significantly associated with an increased risk of CKO.

Regarding kidney outcomes, only SDH and ISH were associated with an increased risk. For 3P-MACE, all BP phenotypes were associated with an increased risk. When the 2024 ESC hypertension criteria ($\geq 140/90$ mmHg) were applied, a clear association was identified between ILDBP and CKO risk. Subgroup analyses revealed that the risk of ILDBP was consistently amplified across all outcomes in participants with diabetes.

Our findings highlight the clinical significance of ISH as a consistent and independent predictor of cardiovascular events. ISH is predominantly observed in older adults and is a well-established risk factor for adverse cardiovascular outcomes in this population.^{6,18} Notably, recent studies have expanded the prognostic relevance of ISH in younger individuals, indicating that its clinical importance is not limited to older populations. In the Chicago Heart Association Detection Project in Industry study, Yano et al¹⁹ reported that, among 27 081 young to middle-aged adults followed for 31 years, ISH (defined as SBP ≥ 140 mmHg and DBP < 90 mmHg) was associated with an increased risk of CVD and cardiovascular mortality compared with optimal-normal BP. Similarly, a Korean nationwide cohort study by Lee et al⁹ involving 6 424 090 adults aged 20 to 39 years found that stage

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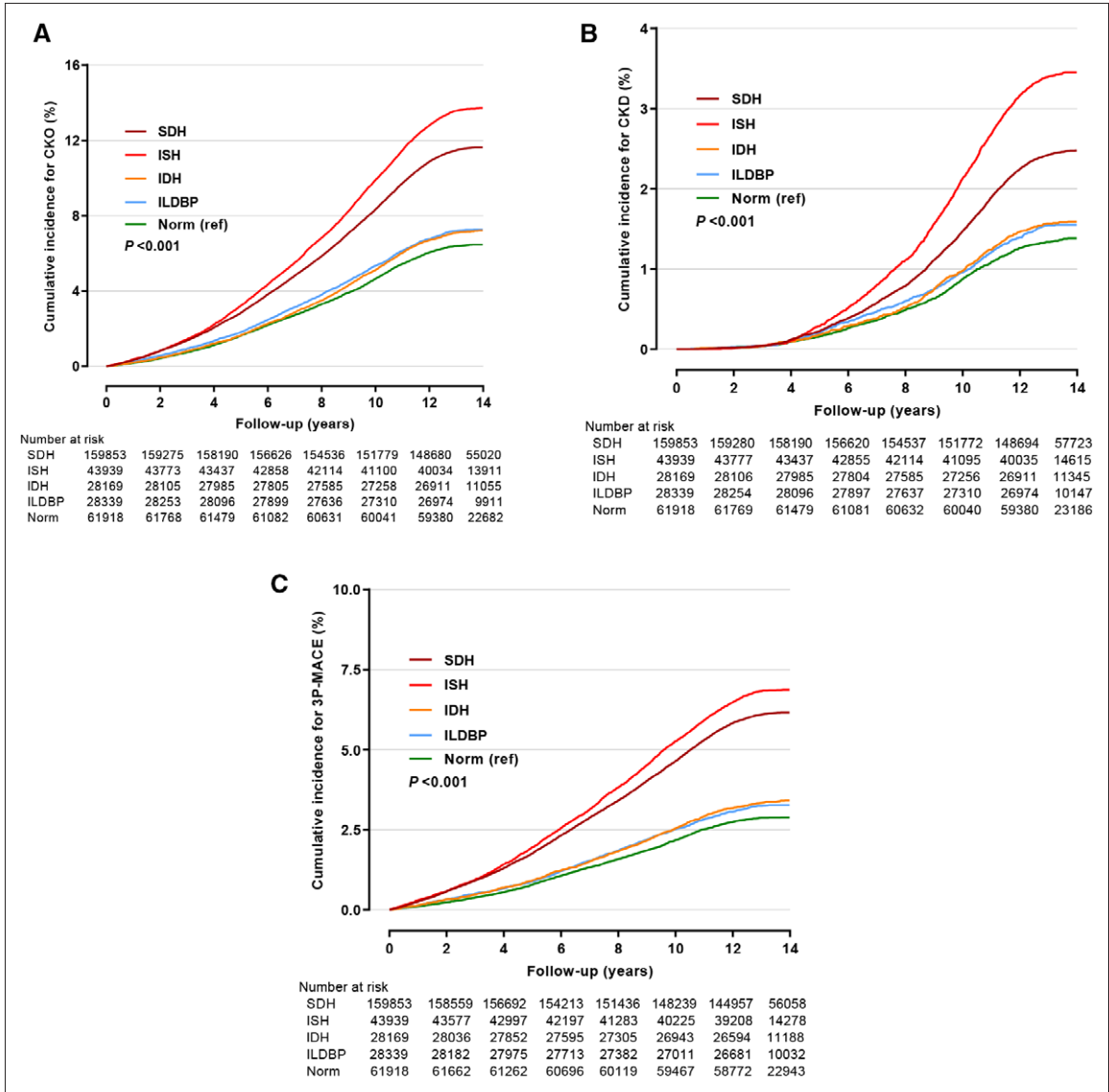


Figure 3. Kaplan-Meier curves for cumulative incidence of (A) CKO, (B) kidney outcome, and (C) 3-point major adverse cardiovascular events (3P-MACE) according to blood pressure (BP) phenotype. BP phenotypes were classified using the 2017 American College of Cardiology/American Heart Association criteria: normotension (Norm; <130/<80 mm Hg), systolic and diastolic hypertension (SDH; ≥130/≥80 mm Hg), isolated systolic hypertension (ISH; ≥130/<80 mm Hg), isolated diastolic hypertension (IDH; <130/≥80 mm Hg), and isolated low diastolic BP (ILDBP; <130/<70 mm Hg). *P* values were calculated using the log-rank test. CKO indicates cardio-kidney outcomes; and PY, person-years.

1 ISH and SDH, defined by the 130/80 mm Hg threshold, were significantly associated with elevated CVD risk. Consistent with these findings, our analysis showed that ISH was associated with an increased risk of 3P-MACE in both younger (<60 years) and older (≥60 years) individuals. Notably, the magnitude of this association was most pronounced among participants aged ≥70 years, with significant effect modification by age. In older subgroups, ISH was associated with a risk at least comparable to that of SDH, potentially reflecting the contribution of age-related arterial stiffness, wider pulse pressure, and reduced vascular compliance. Supporting this, Psaty et al²⁰ demonstrated in the Cardiovascular Health Study that ISH was strongly associated with various measures

of subclinical CVD in a cohort of 5201 community-dwelling older adults. Our study adds to existing evidence by demonstrating that ISH is associated with an increased risk of kidney-specific outcomes, with a magnitude of risk comparable to or even exceeding that of SDH. Few studies have evaluated the association between BP phenotypes and kidney outcomes. A Korean nationwide health-screening database study of 3 030 884 young adults aged 20 to 39 years found that ISH, although associated with a lower risk than SDH, was significantly associated with incident CKD.²¹ A previous cross-sectional study of 1567 middle-aged (45–64 years) adults found that ISH and elevated pulse pressure were associated with the development

of microalbuminuria, a marker of early kidney damage.²² However, to our knowledge, no longitudinal studies have evaluated the association between ISH and incident CKD in middle-aged populations. In a separate study of patients with established CKD, the prevalence of ISH increased with advancing CKD stage, suggesting that worsening kidney function may contribute to the development of ISH and, in turn, amplify cardiovascular risk.²³ This finding underscores the bidirectional nature of the kidney-cardiovascular axis. In this context, our results provide novel evidence that ISH significantly contributes to adverse kidney outcomes, highlighting the importance of early recognition and management of ISH for both cardiovascular and kidney protection.

However, the association between IDH and cardiovascular risk remains controversial. In a cross-sectional analysis of NHANES (2013–2016) and a longitudinal analysis from the Atherosclerosis Risk in Communities study, McEvoy et al²⁴ reported that the estimated prevalence of IDH was substantially higher when defined by the 2017 ACC/AHA criteria (SBP <130 mmHg and DBP ≥80 mmHg) than that with the JNC7 criteria (SBP <140 mmHg and DBP ≥90 mmHg). However, IDH was not significantly associated with an increased risk of cardiovascular events or incident CKD. Similarly, in the Multiethnic Study of Atherosclerosis, involving 5104 adults aged 45 to 84 years, IDH (defined by the 2017 ACC/AHA guideline) was not associated with subclinical CVD, incident systolic hypertension, or future cardiovascular events.²⁵ In contrast, a study using UK Biobank data from 151 831 individuals found that IDH was associated with increased cardiovascular risk only when defined using the 140/90 mmHg threshold, despite a higher prevalence when using the 130/80 mmHg definition.²⁶ However, data from 1.3 million adults in the Kaiser Permanente Northern California cohort showed that IDH was associated with elevated cardiovascular risk regardless of the definition used (≥140/90 mmHg or ≥130/80 mmHg).²⁷ In our study, IDH was also associated with an increased risk of 3P-MACE, but this association was not observed in older individuals. This age-specific difference may reflect variations in the underlying hemodynamic profile, as IDH is more prevalent in younger individuals and may represent early vascular dysfunction. These findings support previous evidence suggesting that IDH may be clinically relevant in younger populations.^{9,21,28} However, IDH was not associated with adverse kidney outcomes in our analysis, indicating that its effect may be confined to cardiovascular rather than cardiorenal risks, especially in middle-aged to older adults. The age-dependent variability in CKO highlights the need for further longitudinal studies to determine whether IDH is a benign early hemodynamic change or an indicator of future disease.

In our study, ILDBP, defined as a low DBP despite normal SBP, was associated with an increased risk of CKO. This finding aligns with the results of a secondary

analysis of the SPRINT trial, which reported that among patients with normalized SBP, an excessively low DBP (<60 mmHg) was associated with an increased risk of major cardiovascular events.^{29,30} The DBP threshold used to define ILDBP in this study was informed by both our findings and prior evidence. In our cohort, although DBP <60 mmHg was associated with increased CKO risk, this category comprised a relatively small subgroup, whereas DBP levels below 70 mmHg were more prevalent and demonstrated a consistent and statistically significant increase in adjusted risk. This threshold is also supported by previous studies showing that DBP levels between 60 and 69 mmHg are already associated with subclinical myocardial injury and adverse cardiovascular outcomes, and that maintaining DBP ≥70 mmHg may be prudent when targeting lower SBP levels.^{1–3} Accordingly, DBP <70 mmHg represents a clinically meaningful cutoff that balances biological relevance, statistical robustness, and generalizability in a population-based setting. Notably, the risk of kidney outcomes associated with ILDBP was particularly pronounced among individuals with diabetes. Although ILDBP was not significantly associated with CKD in the overall population, its adverse effect in patients with diabetes may reflect impaired renal autoregulation owing to autonomic dysfunction, microvascular disease, and arterial stiffness, which increase susceptibility to renal hypoperfusion at low diastolic pressures. In addition, ILDBP was associated with kidney outcome risks when BP phenotypes were defined using the 140/90 mmHg threshold. This finding may help explain the conflicting results across previous studies, where the clinical relevance of low DBP often varied depending on the reference group used rather than the intrinsic harm of low DBP itself. These findings underscore the need for further research to better define the clinical significance of low DBP levels, particularly in high-risk subgroups.

This study has several important strengths. First, by leveraging a large, population-based cohort with long-term follow-up, we were able to systematically evaluate the independent associations of multiple BP phenotypes with both cardiovascular and kidney outcomes. In contrast to prior observational studies that focused on isolated BP components or single outcomes, our unified analytic framework simultaneously compared distinct BP phenotypes using contemporary guideline-based definitions. In addition, the consistency of our findings across different BP thresholds further supports the robustness and clinical relevance of our results. Together, these study features allow a more comprehensive and clinically meaningful assessment of cardio-kidney risk beyond absolute BP values.

Despite the aforementioned strengths, several limitations merit discussion. First, BP was measured at a single time point, which limited our ability to account for BP variability and longitudinal changes. Additionally, ambulatory

BP monitoring, which better captures circadian BP patterns, was not available. Furthermore, changes in BP levels and antihypertensive treatment patterns, including treatment initiation or intensification over time, could not be fully accounted for. Although we adjusted for baseline use of antihypertensive medications, residual confounding related to time-varying BP and treatment exposure cannot be excluded. Second, although we adjusted for numerous covariates, residual confounding factors could not be excluded, particularly with respect to unmeasured variables such as medication adherence, dietary factors, and genetic predisposition. In addition, because our analyses were limited to participants with complete data on all covariates, selection bias related to missing data cannot be excluded. Third, incident CKD was identified using administrative diagnostic and procedure codes, which may underestimate early or subclinical disease and may not precisely reflect the true biological onset of CKD. However, in the UK Biobank, CKD outcomes ascertained using linked hospital inpatient records and procedure codes (*International Classification of Diseases*, Tenth Revision, and Office of Population Censuses and Surveys Classification of Interventions and Procedures, version 4) have been widely used and validated in prior studies, particularly for identifying clinically meaningful and advanced kidney outcomes such as dialysis, kidney transplantation, and vascular access procedures.^{31,32} Accordingly, these code-based definitions are considered highly specific, and any potential misclassification is likely to be nondifferential with respect to BP phenotypes. Fourth, the UK Biobank population was restricted to adults aged 40 to 69 years at baseline, which may limit the generalizability of our findings to younger individuals, among whom IDH is more common. Indeed, the substantial attenuation of risk estimates for SDH and ISH after adjustment for age and sex underscores the strong demographic dependence of these BP phenotypes, particularly their higher prevalence among older individuals and men. Nevertheless, the persistence of statistically significant associations after full multivariable adjustment indicates that SDH and ISH confer cardio-kidney risk beyond that attributable to age and sex alone, with relatively modest additional attenuation after adjustment for metabolic and clinical factors supporting their independent prognostic relevance. Importantly, subgroup analyses further demonstrated significant effect modification by age, but not by sex. The associations between SDH, ISH, and CKO were more pronounced in participants aged ≥ 60 and ≥ 70 years, whereas no meaningful heterogeneity was observed across sex strata. Together, these findings suggest that age functions not only as a major confounder but also as an effect modifier of BP phenotype-related risk, while sex primarily acts as a confounding factor without modifying the strength of these associations. Finally, while our study categorized BP patterns based on established thresholds from the 2017

AHA and 2024 ESC guidelines, the selection of a reference BP group and the specific cutoffs used to define the BP categories can influence both the magnitude and statistical significance of the observed associations. Given that individuals with normal BP by 1 guideline may be classified differently under another guideline, future studies should explore how alternative reference definitions affect the comparative risks associated with distinct BP phenotypes. Relatedly, an important consideration in interpreting our findings is that participants in the UK Biobank were not treated according to the 2017 ACC/AHA hypertension guideline during the study period, as clinical practice largely reflected earlier BP thresholds (SBP ≥ 140 mmHg and DBP ≥ 90 mmHg), including those recommended by the JNC 7 and the 2007 ESC/ESH guidelines.^{33,34} Consequently, individuals classified as hypertensive under the 2017 ACC/AHA definition may not have received antihypertensive treatment or treatment intensification at baseline. However, our study was designed to evaluate the prognostic significance of BP phenotypes rather than the effects of guideline-directed therapy. As such, the observed associations reflect the natural history and long-term risk associated with distinct BP patterns, independent of contemporary treatment strategies. From this perspective, our findings may highlight subgroups who would be considered at elevated risk under current guidelines but may have been undertreated in earlier clinical practice, thereby reinforcing the clinical relevance of BP phenotype-based risk stratification in the context of evolving hypertension guidelines.

In conclusion, this study demonstrates independent and variable associations between distinct BP phenotypes and CKO risk. ISH was associated with an increased risk of adverse kidney outcomes and 3P-MACE, while IDH was associated with a higher risk of 3P-MACE across age groups. ILDBP conferred additional CKD risk among individuals with diabetes. These findings support the need for more refined BP classification systems in clinical practice and indicate that low DBP levels should be managed with particular caution in high-risk populations. Further interventional studies are required to assess whether phenotype-guided BP management strategies can improve CKO.

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Author Contributions

Drs Cho and Park contributed to the research idea and study design. Dr Jhee was responsible for data acquisition. Drs Cho and Ho Jhee contributed to data analysis/interpretation. Drs Park, Cho, and Hyun Jhee performed the statistical analysis, and Dr Hyun Jhee was the guarantor. Each author contributed important intellectual content during article drafting or revision and accepted accountability for the overall work by ensuring that questions pertaining to the accuracy or integrity of any portion of the work were appropriately investigated and resolved.

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Disclosures

None.

Supplemental Material

Tables S1–S9

Figure S1

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